

- 1 A toy car collides into a fixed wall and rebounds with the same speed as shown in Fig. 1.1.

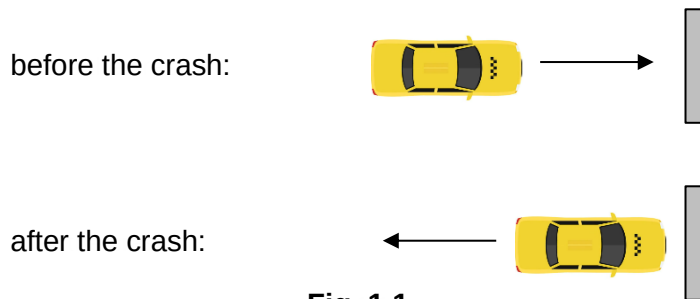


Fig. 1.1

- (a) (i) State the type of collision that occurred between the car and the wall.

..... [1]

- (ii) During the collision the momentum of the car changes. State and explain whether momentum is conserved in this collision.

.....
.....
..... [2]

- (b) Fig. 1.2 shows the variation with time of the magnitude of force acting on the toy car due to the wall during the collision. Take rightwards as positive.

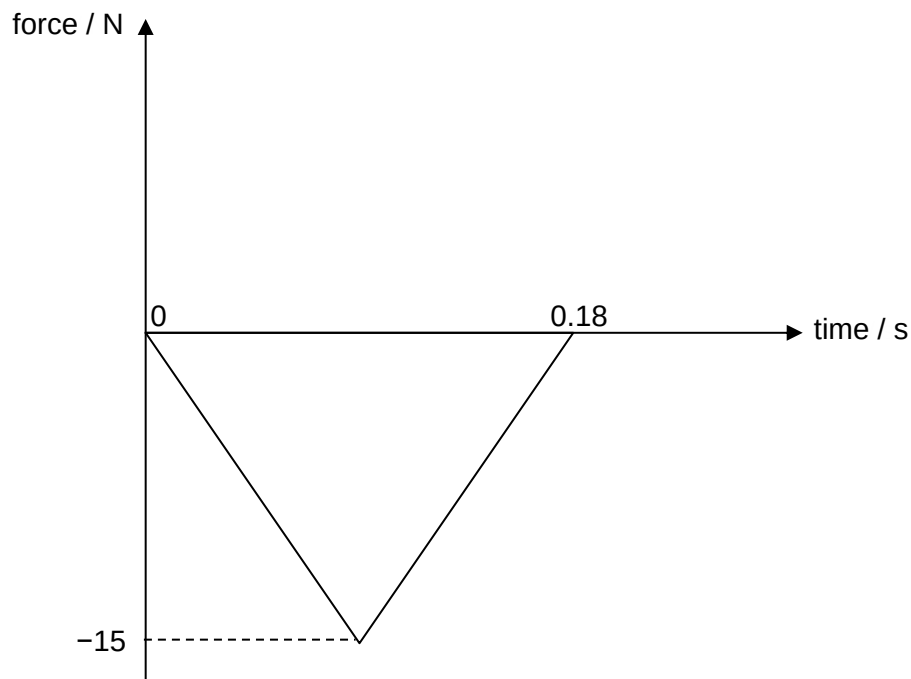


Fig. 1.2

- (i) Determine the magnitude of the initial momentum of the car.

initial momentum = kg m s^{-1} [2]

- (ii) The mass of the toy car is 0.65 kg. Determine the magnitude of average acceleration of the toy car during the collision.

average acceleration = m s^{-2} [2]

- (iii) A rubber bumper is now secured to the front of the toy car. The toy car undergoes a similar collision with the wall.

On Fig. 1.2, sketch a new graph showing the variation with time of the force acting on the toy car.

[1]